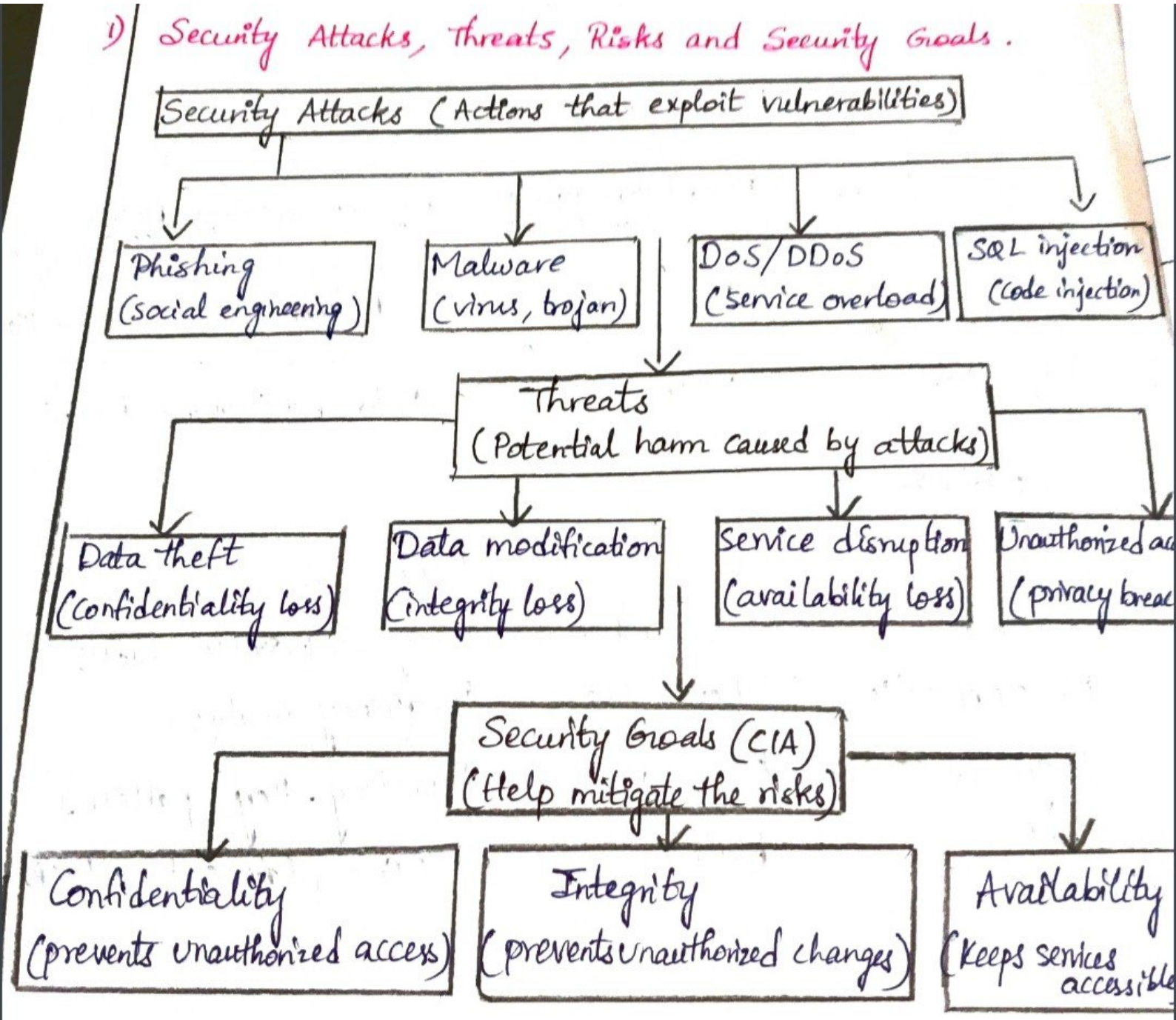
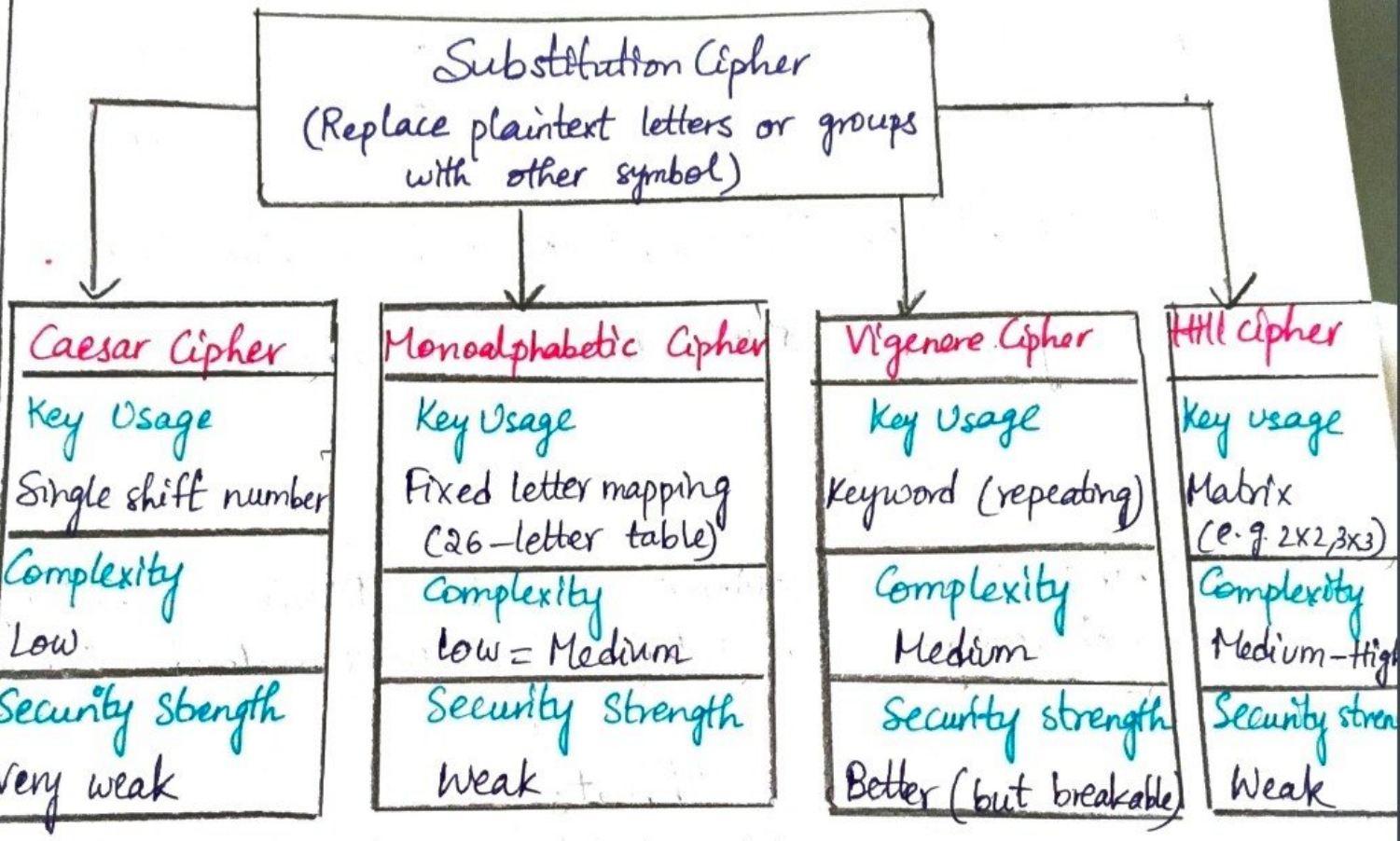
1) Security Attacks, Threats, Risks and Security Goals.



Real-world Example for each link

Attack → Threat → Risk	Security Goal (mitigation)	Real-life Exam
Phishing → data theft → financial loss	Use MFA to prevent unauthorized access	Fake bank emai leading to accou Compromise.

2)



Comparison

Technique	Key Usage	Main Weakness
Caesar	Shift number	Only 25 possible shifts
Monoalphabetic	letter table	Frequency analysis
Vigenere	Keyword	Repeating-Key patterns
Hill	Matrix	Known-plaintext attack

3)

Classical Encryption Schemes
(Early Techniques to protect data)

Substitution Techniques
(Change the Identity of letters)

Examples
* Caesar Cipher
* Monoalphabetic
* Vigenere
* Hill

Key feature
* Alters letter values
* Keeps letter order (usually)

Transposition Techniques
(Change the position of letters)

Examples
* Rail fence cipher
* Columnar Transposition
* Double transposition

Key feature
* Rearranges order
* Keeps let identity

Comparison

Aspect	Substitution	Transposition
Operation	Replaces letters	Rearranges letters
Data structure	Letter positions unchanged	Letter order changes
Security level	Weak	Weak
Example	Caesar cipher	Rail fence cipher

4)

Steganography
(Hides secret information inside an original file/message)

Data Hiding
(Conceals secret data in media)

What it hides – Secret information

Media – Image, audio, video, text

Example – Text hidden in an image

Watermarking
(Shows ownership/authencity)

What it adds – Visible

Media – Image, audio, video, document

Example – Copyright watermark on a photo.

Survivability
(Keeps data/watermark intact)

What it ensures – Survives changes

Media – Image, vide, audi

Example – Waterma remains after compress

Secure Communication

Data Hiding
Keeps the message secret from unauthorized users.

Water marking
Helps in ownership, authencity and content tracking.

Survivability
Ensures the hidden remains detectable eve after modification

eal-world Applications

Data Hiding
ecret text in an image

Watermarking
Copyright mark on a video/photo.

Survivability
Robust watermark in media files